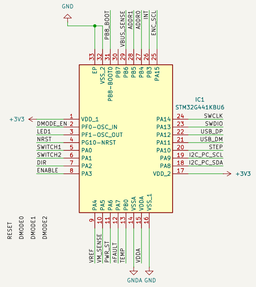
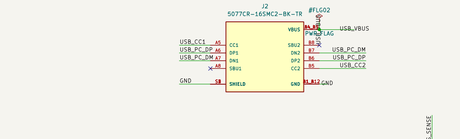


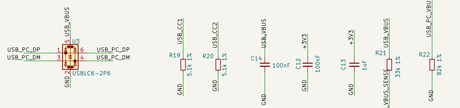




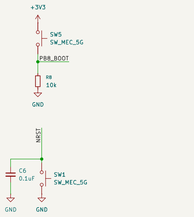
TERM: POPULATE AT BUS ENDS ONLY

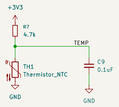


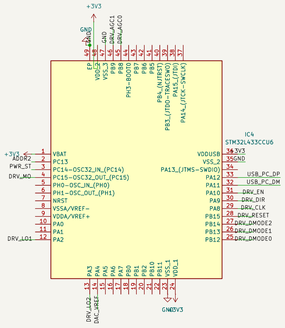

USB FS DEVICE / SELF-POWERED
USB_VBUS isolated from board +5V (no backfeed).
PA9/_SPIO VBUS detect; enable USB only with VBUS.
VDDUSB = +3V3; enable PWR_M2_SV1 in firmware.
Internal D+ pull-up / termination. 90-ohm differential PCB pair.





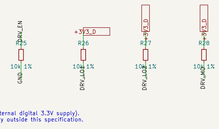




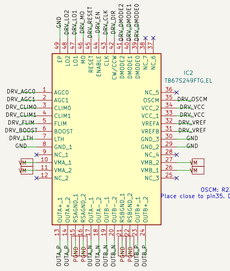







LO1/LO2/MO pull-ups: +3V3_D only (external digital 3.3V supply).
VM and +3V3_D supplied by power circuitry outside this specification.







VREF: R29=10k series, R30=9.1k to GND. C21=100nF.
3.3V DAC full scale -> VREF 1.57V -> IOUT 1.97A.
R30/C21 GND return near pins29/30, away from OUT/VM.










citors: C15 bulk 220uF/63V: C16 at VMA: C17 at VMB: C18 10uF/50V.
Keep VM-PGND loops short: paired pins joined at IC on PCB.

ngle PGND-GND copper connection directly beneath IC2 exposed pad.
ds to GND, dense thermal vias: 4 layers / 2oz. Motor outputs: 5A routing.

er DAC1_OUT1 PA4 buffer enabled: reference VDDA/VREF+ (+3V3_A).
d DAC may retain ~0.2V. ENABLE Low or disable DAC for zero current.
eck VM -> set DAC -> set DMODE -> wait >=7.5us -> ENABLE High.
: ENABLE/DMODE/AGC Low: evaluate initially with AGC0=AGC1=Low.